

# CDS Agent Clinical Knowledge Base

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## Diabetes Management

### Overview

Diabetes Mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels. Patients with diabetes are at increased risk of cardiovascular disease, chronic kidney disease, neuropathy, retinopathy, and lower extremity complications. Effective diabetes management requires continuous monitoring, preventive screening, and timely intervention.

### Clinical Decision Support Rules

**HbA1c Monitoring:** Verify whether an HbA1c test has been performed within the last 12 months. If no recent result is available, recommend ordering an HbA1c test.

**Diabetic Eye Examination:** Determine whether the patient has completed an annual retinal eye examination. If not, generate a retinal screening recommendation.

**Kidney Disease Monitoring:** Review urine albumin and eGFR results. If kidney monitoring has not been completed within the last year, recommend renal screening.

**Diabetic Foot Examination:** Confirm annual foot assessment completion. Missing documentation should trigger a preventive care reminder.

**Uncontrolled Diabetes Detection:** Patients with HbA1c values above 9% should be flagged as potentially having uncontrolled diabetes and medication optimization should be considered.

## Hypertension Management

### Overview

Hypertension is a major contributor to cardiovascular disease, stroke, and kidney disease. Early identification and regular blood pressure monitoring are essential.

**Elevated Blood Pressure:** If BP exceeds 140/90 mmHg, recommend reviewing the treatment plan.

**Missing Blood Pressure Documentation:** If no BP measurement exists during the encounter, prompt staff to capture vitals.

**Follow-Up Compliance:** Patients without a follow-up visit within six months should receive a follow-up recommendation.

## Heart Failure Management

### Overview

Heart failure is a chronic condition associated with high morbidity and hospitalization rates.

Ejection Fraction Documentation: Missing EF should trigger an echocardiogram review recommendation.

Fluid Retention Monitoring: Rapid weight gain may indicate fluid retention and requires evaluation.

Specialist Follow-Up: Recommend cardiology consultation if no recent review exists.

## Chronic Kidney Disease (CKD) Management

### Overview

CKD requires regular monitoring of kidney function and early intervention.

eGFR Monitoring: Missing eGFR results should trigger testing recommendations.

Creatinine Monitoring: Missing recent creatinine results should trigger repeat testing.

Advanced CKD Referral: Stage 4 or 5 CKD patients should be evaluated for nephrology referral.

## COPD Management

### Overview

COPD is a progressive respiratory disease often associated with tobacco exposure.

Smoking Status Evaluation: Current smokers should receive smoking cessation counseling.

Spirometry Assessment: Missing spirometry documentation should trigger evaluation.

Vaccination Monitoring: Review influenza vaccination status regularly.

## Asthma Management

### Overview

Asthma management focuses on symptom control and prevention of exacerbations.

Rescue Inhaler Utilization: Excessive use may indicate poor asthma control.

Asthma Action Plan Verification: Patients without documented action plans should receive recommendations.

## **Preventive Care and Screening**

Mammography Screening: Women aged 50 to 74 should be evaluated for mammogram compliance.

Colorectal Cancer Screening: Eligible adults aged 45 to 75 should be assessed.

Lung Cancer Screening: Smokers should be evaluated for eligibility.

Prostate Cancer Screening Discussion: Eligible male patients should receive PSA discussion reminders.

## **Vaccination Management**

Influenza Vaccination: Assess annually.

Pneumococcal Vaccination: Evaluate adults aged 65+.

Shingles Vaccination: Screen adults aged 50+.

## **Medication Safety Monitoring**

Drug Interaction Detection: Review active medications and identify significant interactions.

Allergy Verification: Compare medications against documented allergies and trigger alerts for conflicts.

## **Mental Health Management**

Depression Assessment: Ensure PHQ-9 assessments are documented.

Behavioral Health Follow-Up: Generate reminders for overdue follow-ups.